

Tharaka A. Jayalath

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Professional Summary

Environmental and resource economist with a Ph.D. and seven years of quantitative research spanning water, agriculture, and environmental policy. Work combines discrete choice modeling, hedonic pricing, causal inference, and ML-based crop simulation in R and Python across datasets reaching 3.5 million records. Published in *Nature* (2026), *Environmental and Resource Economics* (2023), *Canadian Water Resources Journal* (2025), and *Ecosystem Services* (2021).

Education

Ph.D. in Applied Economics	2020 – 2025
University of Saskatchewan, Saskatchewan, Canada	
M.Sc. in Economics	2016 – 2018
Mississippi State University, Mississippi, United States	
Minor in Geospatial Information Systems and Remote Sensing	
B.Sc. in Agricultural Economics (Hons)	2008 – 2012
Wayamba University, Sri Lanka	

Certifications

The Data Science of Health Informatics · Johns Hopkins University	2026
Google Data Analytics · Google	2026
Agentic AI Builder's Bootcamp · Maven	2026
AI Engineering Bootcamp: Build AI Agents with Claude Code, RAG and MCP · Maven	2026

Research Experience

Postdoctoral Fellow, University of Alabama 2026 – Present

- Design and deploy ETL workflows across heterogeneous administrative and environmental databases, applying join-level quality control and version control to produce analysis-ready datasets.
- Compiled USDA NASS county-level data, commodity price series, and scraped retail indicators into multi-source economic datasets.
- Analyze multi-dimensional water, weather, and climate datasets using statistical and machine learning models; deliver quantitative outputs for federal agency program evaluation.
- Translate statistical model outputs into structured technical reports and decision-ready formats for non-technical federal stakeholders.
- Built a time series classification pipeline to detect onset and severity patterns across multi-source climate and satellite data streams.
- Built an image processing pipeline to extract spectral indices and land cover classifications from multi-temporal raster data at scale.
- Developed Quarto and Tableau visualizations translating large-scale model outputs into interactive reports for federal program stakeholders.
- Structured Python and R pipelines with version control and automated QC to ensure reproducible estimation across heterogeneous data sources.
- Established data access agreements with external agencies and research groups, specifying data elements and transfer format; coordinated data delivery under formal governance protocols.

Graduate Research Assistant, University of Saskatchewan 2020 – 2025

- Designed 4-version split-sample SP survey (n=3,850) with GIS-assigned status quo to isolate causal effects of spatial context on choice.

- Estimated mixed logit models combining stated and revealed preference data, producing use and non-use values for environmental cost-benefit analysis.
- Extended to Random Parameters Logit (RPL) with simulation-based MLE to generate respondent-level WTP distributions.
- Quantified water quality capitalization via hedonic fixed-effects regression (326,191 Ontario transactions) integrating Statistics Canada Census dissemination area boundaries and CanVec lake polygons in R.
- Built a Travel Cost Method (TCM) model on 3.5 million citizen-science records to estimate recreational demand and inform natural resource policy.
- Estimated average and marginal irrigation shadow prices via AquaCrop-OS simulation (397 Saskatchewan fields) under climate uncertainty.
- Synthesized 29 hedonic studies via meta-regression and geospatial benefit transfer; national water-quality estimate \$8.26B.
- Co-investigated a \$200,000 Agriculture Development Fund grant; authored four peer-reviewed manuscripts.
- Established data access agreements with external agencies, specifying data elements and transfer format; managed data acquisition under REB-approved governance protocols.

Graduate Research Assistant, *Mississippi State University*

2016 – 2018

- Administered contingent valuation survey (n=242); joined U.S. Census demographic covariates and USGS shapefiles via ArcGIS to model enrollment.
- Estimated landowner WTA for USDA grassland enrollment (probit, n=242); mean WTA \$290/acre/yr; total cost \$3B/yr = 57x 2016 CRP.
- Instructed undergraduate and graduate students in timber investment analysis and forest valuation methods (Teaching Assistant, Fall 2017).

Peer-Reviewed Publications

- [5] Jayalath, T., Lloyd-Smith, P., & Richard, V. (2024). Valuing local water quality in Ontario using revealed preference data. *Canadian Journal of Agricultural Economics*. In press.
- [4] Jayalath, T., & Lloyd-Smith, P. (2025). Water quality and property values: A meta-analysis and Canadian national benefit assessment. *Canadian Water Resources Journal*, 50, 101303.
doi:10.1080/07011784.2025.2543860
- [3] Jayalath, T., Lloyd-Smith, P., & Becker, M. (2023). Biodiversity benefits of birdwatching using citizen science data and individualized recreational demand models. *Environmental and Resource Economics*, 86(1), 83–107.
doi:10.1007/s10640-023-00788-0
- [2] Jayalath, T., Grala, R. K., Grado, S., & Evans, D. (2021). Increasing provision of ecosystem services through participation in a conservation program. *Ecosystem Services*, 50, 101303.
doi:10.1016/j.ecoser.2021.101303
- [1] Brodeur, A., Mikola, D., Cook, N., ..., & Jayalath, T. (2026). Reproducibility and robustness of economics and political science research. *Nature*, 652(8108), 151–156.
doi:10.1038/s41586-026-10251-x

Manuscripts Under Review

- [2] Jayalath, T., & Lloyd-Smith, P. (2025). The economic value of irrigation water under policy and environmental changes. *Under review, Journal of Water Economics*.
- [1] Jayalath, T., & Lloyd-Smith, P. (2025). Varying status quo conditions in stated preference surveys: Empirical evidence for water quality. *Manuscript in preparation*.

Technical Skills

Languages	Python, R, SQL, Stata, MATLAB
Tools	FastAPI, Streamlit, Pinecone, Databricks, AquaCrop-OS, ArcGIS, QGIS, Quarto, Tableau, $\LaTeX$, Excel
Methods	Machine Learning, Discrete Choice Modeling (MNL / Mixed Logit / RPL), Causal Inference, Hedonic Pricing, RAG Pipeline Development, AI Agent Design, Stochastic Simulation, Meta-Analysis, Geospatial Analysis
Applied	Non-market valuation, agricultural systems modeling, water security analysis, environmental cost-benefit, AI agent workflows, policy communication

Conference & Invited Presentations

- Annual Meeting of the Canadian Resource and Environmental Economics Association (CREEA), Kelowna, British Columbia, Canada (2024)
- Annual Meeting of the Canadian Agricultural Economics Society (Virtual, 2022)
- National Convention, Society of American Foresters (Virtual, 2021)
- Annual Conference of the International Society of Forest Resource Economics, New Orleans, USA (2017)

Honors & Awards

Mickleborough Graduate Scholarship in Agricultural Economics (USASK)	2023
The Red Williams Post-Graduate Scholarship (USASK)	2022
Dollie Hantelman Agriculture Scholarship (USASK)	2021
Graduate Research Assistantship Award (USASK)	2020
Graduate Research Assistantship Award (MSU)	2017

Grants & Funding

- Agriculture Development Fund (Saskatchewan Ministry of Agriculture, Grant #20230060, 2023–2026). *Economic Modelling of Irrigation Decision-Making for Sustainable Agricultural Water Management*. \$200,000. Role: Research Economist.

Professional Service

Peer Reviewer:

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| • Land Use Policy | • Environmental and Resource Economics |
| • World Leisure Journal | • Water Economics and Policy |